

Comparison of Tikhonov and Total Variation Regularization for Noisy Linearized Electrical Impedance Tomography

Name: Xi Chen Student ID: u7641463
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ABSTRACT

Electrical Impedance Tomography (EIT) is a severely ill-posed inverse problem in which stable conductivity reconstruction requires regularization. Tikhonov and Total Variation (TV) regularization are compared within a linearized EIT framework using synthetic finite element data generated from both discontinuous and smooth conductivity phantoms under noise levels ranging from 0% to 10%. TV regularization provides more accurate reconstructions for discontinuous conductivities by preserving sharp interfaces, whereas Tikhonov regularization is generally more effective for smooth conductivity distributions when the regularization terms are active. The results show that reconstruction quality depends strongly on both the regularization parameter regime and the compatibility between the regularization model and the underlying conductivity structure. Significant reconstruction errors persist even for noise-free data, indicating that linearization, discretization, and regularization bias can be comparable to or exceed the effects of measurement noise. These findings highlight the importance of prior selection in EIT reconstruction and reveal fundamental limitations of the linearized inverse problem.

1 INTRODUCTION

Electrical Impedance Tomography (EIT) is a non-invasive imaging technique in which electrical currents are applied on the boundary of a body and the resulting boundary voltages are used to infer the interior conductivity distribution. Since the measurements are indirect, limited to the boundary, and highly sensitive to perturbations, EIT reconstruction is a severely ill-posed inverse problem and requires regularization to obtain stable solutions.

This paper investigates the influence of regularization on linearized EIT reconstruction. In particular, Tikhonov regularization, which promotes smooth conductivity distributions, is compared with Total Variation (TV) regularization, which favours piecewise-constant solutions and sharp interfaces. To examine the suitability of these competing priors, two conductivity phantoms are considered: a discontinuous circular inclusion and a smooth Gaussian inclusion. Reconstructions are computed under noise levels ranging from 0% to 10% in order to assess both reconstruction accuracy and robustness to measurement noise.

The numerical experiments show that reconstruction quality depends strongly on the compatibility between the regularization model and the underlying conductivity structure. TV regularization is more effective for discontinuous conductivities, whereas Tikhonov regularization is generally more suitable for smooth conductivity distributions when the regularization terms are active. The results also demon-

strate a strong dependence on the regularization parameter regime and reveal that substantial reconstruction errors persist even for noise-free data, indicating that modelling error can be as significant as measurement noise in linearized EIT. These findings highlight both the importance of prior selection and the intrinsic limitations of the linearized reconstruction framework.

2 BACKGROUND

2.1 Electrical Impedance Tomography

The mathematical foundations of EIT can be traced to Calderón's seminal work on the inverse conductivity problem [1], which asks whether the electrical conductivity inside a domain can be determined from measurements made only on its boundary. This problem has since become a central topic in inverse problems, motivating extensive research into questions of uniqueness, stability, and numerical reconstruction. Despite significant theoretical advances, practical EIT reconstruction remains challenging due to the limited and indirect nature of boundary measurements.

2.2 Ill-posedness and Regularization

The inverse conductivity problem in EIT is severely ill-posed, meaning that small errors in the measurement data may lead to large errors in the reconstructed conductivity. Since practical measurements are inevitably contaminated by noise, direct inversion is generally unstable. Regularization is therefore required to obtain meaningful reconstructions.

Two of the most widely used regularization techniques are *Tikhonov regularization* [2] and *Total Variation (TV) regularization* [3]. Tikhonov regularization promotes smooth conductivity distributions and tends to suppress noise-induced oscillations, whereas TV regularization preserves discontinuities and is better suited to recovering sharp conductivity interfaces. The choice of regularization therefore reflects different assumptions about the underlying conductivity structure.

TV regularization has been applied successfully to EIT by Borsic et al. [4], who developed a primal-dual interior-point method for TV-regularized reconstruction. Their results demonstrated improved edge preservation and noise robustness compared with traditional smoothness-based approaches. The numerical method used in this project is based on this framework.

2.3 Motivation

Much of the EIT literature focuses on the development of new reconstruction algorithms [5] [6] or the demon-

stration of existing methods on particular imaging scenarios. Although both Tikhonov and TV regularization have been studied extensively, direct comparisons between these methods under controlled conditions are less common, particularly when the effects of conductivity structure and measurement noise are examined.

The effectiveness of a regularization method depends not only on the measurement data but also on how well its underlying assumptions reflect the true conductivity distribution. A method that performs well for smooth conductivities may be less suitable for recovering sharp interfaces, while a method designed to preserve discontinuities may introduce artefacts when the conductivity varies smoothly. Understanding these trade-offs is essential for selecting appropriate reconstruction techniques in practical EIT applications.

3 THEORY

3.1 Forward Problem and Experimental Setup

The numerical experiments are performed on the square domain $\Omega = [0, 1]^2$. The boundary $\partial\Omega$ is partitioned into sixteen equal-length electrode segments $e_1, \dots, e_{16}$, ordered counterclockwise around the boundary with four electrodes on each side. Current injection follows the adjacent-drive protocol. For the k -th current pattern, unit current is injected through electrode e_k and removed through the adjacent electrode e_{k+1} , with the periodic convention $e_{17} = e_1$. Defining

$$I_l^{(k)} = \begin{cases} 1, & l = k, \\ -1, & l = k + 1, \\ 0, & \text{otherwise,} \end{cases}$$

the corresponding boundary current density is

$$g_k(x) = \sum_{l=1}^{16} \frac{I_l^{(k)}}{|e_l|} \chi_{e_l}(x),$$

which satisfies the compatibility condition $\int_{\partial\Omega} g_k ds = 0$. For each current pattern, the electric potential $u^{(k)}$ satisfies the Neumann problem

$$\begin{cases} \nabla \cdot (\sigma \nabla u^{(k)}) = 0, & x \in \Omega, \\ \sigma \frac{\partial u^{(k)}}{\partial n} = g_k, & x \in \partial\Omega, \end{cases} \quad (1)$$

together with the grounding condition $\int_{\Omega} u^{(k)} dx = 0$.

The forward problem is solved using the finite element method. In weak form, $u^{(k)} \in H^1(\Omega)$ satisfies

$$\int_{\Omega} \sigma \nabla u^{(k)} \cdot \nabla v dx = \int_{\partial\Omega} g_k v ds, \quad \forall v \in H^1(\Omega). \quad (2)$$

The electrode voltages are approximated by the electrode-average potentials

$$U_l^{(k)} = \frac{1}{|e_l|} \int_{e_l} u^{(k)} ds, \quad l = 1, \dots, 16,$$

and the measured data are taken as adjacent voltage differences

$$V_l^{(k)} = U_l^{(k)} - U_{l+1}^{(k)}, \quad U_{17}^{(k)} = U_1^{(k)}.$$

Synthetic EIT data are generated by solving the forward problem for both the homogeneous reference conductivity $\sigma_0 = 1$ and a target conductivity distribution σ . The resulting difference measurements are

$$\Delta V_l^{(k)} = V_l^{(k)}(\sigma) - V_l^{(k)}(\sigma_0).$$

Collecting all 16×16 measurements yields the data vector $\Delta V \in \mathbb{R}^{256}$.

Measurement noise is simulated according to

$$\Delta V^\delta = \Delta V + \delta \frac{\|\Delta V\|_2}{\|\eta\|_2} \eta, \quad \eta \sim N(0, I), \quad (3)$$

where δ denotes the relative noise level. The noisy data ΔV^δ are subsequently used in the linearized inverse problem $Jz \approx \Delta V^\delta$.

3.2 Linearized Inverse Problem

Let $F : \sigma \mapsto V$ denote the nonlinear forward operator mapping the conductivity distribution to the measured boundary voltages. Using the homogeneous reference conductivity $\sigma_0 = 1$, the conductivity is represented as $\sigma = \sigma_0 + \Delta\sigma$ with corresponding voltage difference

$$\begin{aligned} \Delta V &= F(\sigma) - F(\sigma_0) \\ &= F(\sigma_0 + \Delta\sigma) - F(\sigma_0) \\ &= F'(\sigma_0) \Delta\sigma + \mathcal{O}(\|\Delta\sigma\|^2) \approx J \Delta\sigma, \end{aligned} \quad (4)$$

where $J = F'(\sigma_0)$ denotes the sensitivity matrix [7].

Let $\Omega = \bigcup_{n=1}^N q_n$ denote the finite element reconstruction mesh. The conductivity perturbation is approximated by a piecewise-constant function, $\Delta\sigma(x) = \sum_{n=1}^N z_n \chi_{q_n}(x)$, where χ_{q_n} is the indicator function of the element q_n and $z = (z_1, \dots, z_N)^T$ contains the unknown conductivity coefficients. Substituting this representation into (4) yields

$$Jz \approx \Delta V. \quad (5)$$

The entries of the sensitivity matrix are computed from the reference solutions through

$$J_{jk,n} = \int_{q_n} \nabla u_j^{(0)} \cdot \nabla u_k^{(0)} dx, \quad (6)$$

[8] where $u_j^{(0)}$ and $u_k^{(0)}$ are the potentials corresponding to the homogeneous conductivity σ_0 . Replacing the exact voltage difference by the noisy measurements ΔV^δ yields the discrete inverse problem

$$Jz \approx \Delta V^\delta. \quad (7)$$

The matrix J is highly ill-conditioned, making the reconstruction sensitive to measurement noise [1, 7]. Consequently, regularization is required to obtain stable conductivity reconstructions.

3.3 Regularized Reconstruction

The discrete inverse problem $Jz \approx \Delta V^\delta$ is highly ill-posed due to the severe ill-conditioning of the sensitivity matrix J . To obtain stable reconstructions, regularization is introduced through the variational problem [10]

$$z_\alpha = \arg \min_z \left\{ \frac{1}{2} \|Jz - \Delta V^\delta\|_2^2 + \alpha R(z) \right\}, \quad (8)$$

where $\alpha > 0$ is the regularization parameter and $R(z)$ is a penalty functional. For Tikhonov regularization [2],

$$R(z) = \|z\|_2^2,$$

giving

$$z_{\text{Tik}} = \arg \min_z \left\{ \frac{1}{2} \|Jz - \Delta V^\delta\|_2^2 + \alpha \|z\|_2^2 \right\}. \quad (9)$$

Tikhonov regularization promotes smooth conductivity distributions and provides a baseline reconstruction method.

For total variation (TV) regularization [3], $R(z) = \|Dz\|_1$, giving

$$z_{\text{TV}} = \arg \min_z \left\{ \frac{1}{2} \|Jz - \Delta V^\delta\|_2^2 + \alpha \|Dz\|_1 \right\}. \quad (10)$$

TV regularization promotes piecewise-constant reconstructions and is particularly effective for recovering sharp conductivity interfaces.

The TV problem is solved using a primal-dual interior point method (PD-IPM) based on the framework of Borsic et al [4]. The corresponding smoothed optimality system is

$$J^T(Jz - \Delta V^\delta) + \alpha D^T y = 0, \quad (11)$$

$$Dz - Ey = 0, \quad E_i = \sqrt{(Dz)_i^2 + \beta}, \quad (12)$$

where y is the dual variable and $\beta > 0$ is a smoothing parameter.

Modifications introduced in this experiment. The original method of Borsic et al [4]. was developed for nonlinear EIT, requiring repeated forward solves and Jacobian updates throughout the reconstruction process. In contrast, the present work considers the linearized inverse problem $Jz \approx \Delta V^\delta$, for which the sensitivity matrix J is computed once and remains fixed during the iteration. This substantially reduces the computational cost associated with repeated evaluations of the forward model.

A second modification concerns the solution of the primal-dual Newton system. In the original PD-IPM formulation, the Newton step is obtained from a coupled saddle-point system involving both the primal variable z and the dual variable y . For large finite element meshes, the resulting system can become computationally expensive to assemble and solve. To avoid this difficulty, the dual variables are eliminated, yielding a reduced Newton equation for the primal update δz ,

$$\left(J^T J + \alpha D^T W D \right) \delta z = -r_1 - \alpha D^T (E^{-1} r_2), \quad (13)$$

where

$$r_1 = J^T (Jz - \Delta V^\delta) + \alpha D^T y, \quad r_2 = Dz - Ey,$$

and

$$W_i = \frac{1 - \frac{y_i (Dz)_i}{E_i}}{E_i}.$$

The matrix in (13) is symmetric positive definite [9], allowing the Newton step to be computed efficiently without forming the full saddle-point system. Once δz has been obtained, the dual update is recovered separately.

The reduced system is solved using a matrix-free conjugate gradient method [9], requiring only matrix-vector products with J , D , and their transposes. This avoids the explicit construction and storage of large dense matrices, making the algorithm suitable for finer reconstruction meshes.

To improve robustness, a small damping term is added to the reduced system together with diagonal preconditioning. Furthermore, the smoothing parameter β is decreased gradually during the iteration,

$$\beta_{k+1} = \max(\gamma \beta_k, \beta_{\min}), \quad 0 < \gamma < 1,$$

providing a continuation strategy from a smooth approximation of the TV functional towards the nonsmooth ℓ^1 penalty. This continuation improves convergence while preserving the edge-preserving properties of TV regularization.

These modifications retain the primal-dual framework of Borsic et al. while significantly reducing computational cost for the large-scale linearized EIT reconstructions. “latex

3.4 Test Phantoms and Evaluation Criteria

Two conductivity phantoms are considered to investigate the performance of Tikhonov and TV regularization for both discontinuous and smooth conductivity distributions.

Phantom A (sharp inclusion). The first phantom consists of a piecewise-constant circular inclusion,

$$\sigma_A(x) = \begin{cases} 1 + \kappa, & x \in D_1, \\ 1, & x \notin D_1, \end{cases}$$

where $D_1 \subset \Omega$ denotes a circular inclusion. This phantom is used to assess the ability of the reconstruction methods to recover sharp conductivity interfaces.

Phantom B (smooth inclusion). The second phantom is a smooth Gaussian-type conductivity perturbation,

$$\sigma_B(x) = 1 + \kappa \exp(-c|x - x_0|^2),$$

where $x_0 \in \Omega$ is the centre of the inclusion and $c > 0$ controls its width. This phantom is used to evaluate the reconstruction of smoothly varying conductivity distributions.

For each phantom, synthetic voltage measurements are generated using the forward FEM model described in Section 3.1. Reconstructions are computed under several noise levels,

$$0\%, \quad 1\%, \quad 3\%, \quad 5\%, \quad 10\%.$$

The reconstruction quality is quantified using the relative error

$$E_{\text{rel}} = \frac{\|z - z_{\text{true}}\|_2}{\|z_{\text{true}}\|_2}, \quad (14)$$

and is further assessed through visual inspection of the reconstructed conductivity images and their robustness with respect to measurement noise.

Based on the characteristics of the two regularization methods, TV regularization is expected to perform better for Phantom A, where the true conductivity contains a discontinuous interface. Since TV promotes piecewise-constant solutions, it is anticipated to preserve sharp boundaries more effectively than Tikhonov regularization. In contrast, for Phantom B the conductivity varies smoothly, and Tikhonov regularization is expected to provide more accurate reconstructions. TV regularization may introduce staircase-type artefacts due to its preference for piecewise-constant solutions.

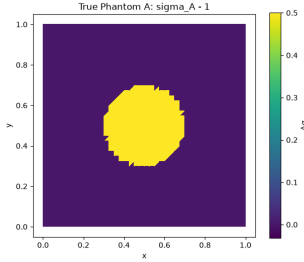
4 RESULTS

4.1 Phantom A

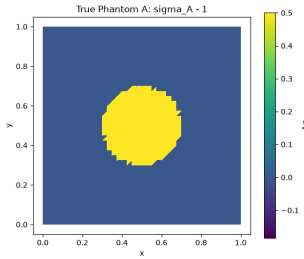
The first test problem consists of a circular conductivity inclusion embedded in a homogeneous background. The conductivity is defined by

$$\sigma_A(x) = \begin{cases} 1 + \kappa, & x \in D_1, \\ 1, & x \notin D_1, \end{cases}$$

with $\kappa = 0.5$.



(a) True conductivity σ_A .



(b) Conductivity perturbation $\Delta\sigma_A = \sigma_A - \sigma_0$.

Figure 1: Ground-truth conductivity distribution and conductivity perturbation for Phantom A.

Since the inverse problem reconstructs $\Delta\sigma_A = \sigma_A - \sigma_0$, the perturbation image in Figure 1b is used as the reference for all subsequent reconstruction comparisons.

We first examine the reconstructions obtained using Tikhonov regularization. Figures 2–6 show the reconstructed conductivity perturbations for Phantom A under increasing noise levels.

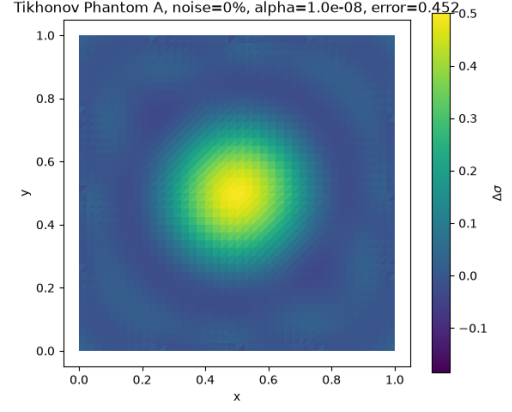


Figure 2: Tikhonov reconstruction of Phantom A for noise level 0%.

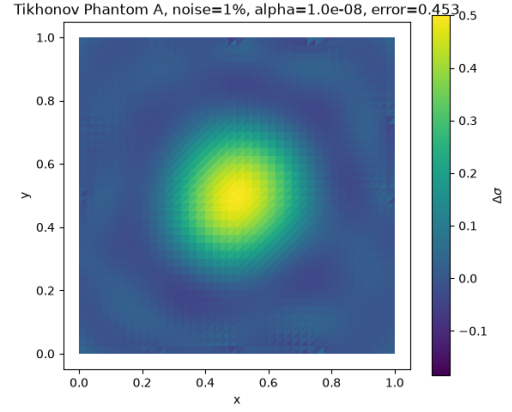


Figure 3: Tikhonov reconstruction of Phantom A for noise level 1%.

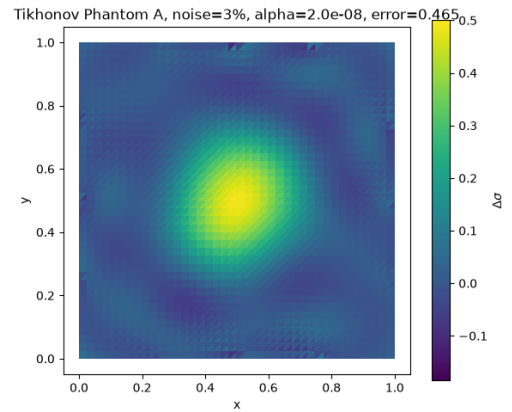


Figure 4: Tikhonov reconstruction of Phantom A for noise level 3%.

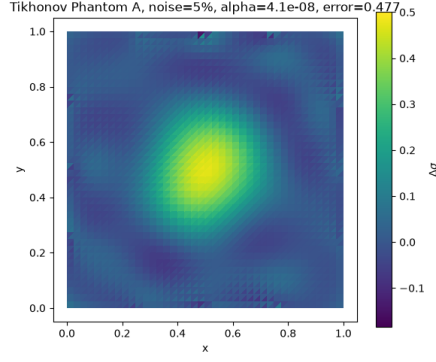


Figure 5: Tikhonov reconstruction of Phantom A for noise level 5%.

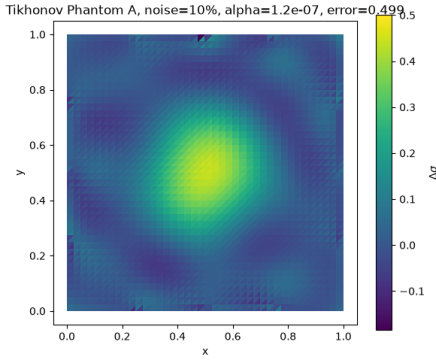


Figure 6: Tikhonov reconstruction of Phantom A for noise level 10%.

Figures 2–6 show that Tikhonov regularization is able to recover the approximate location of the circular inclusion across the tested noise levels. However, the reconstructed inclusion is noticeably smoothed, with a diffuse transition between the inclusion and the background. This behaviour is consistent with the quadratic penalty in Tikhonov regularization, which stabilizes the inverse problem by suppressing oscillations but also tends to blur sharp interfaces. As the noise level increases, the reconstructed contrast becomes less distinct and the boundary of the inclusion becomes harder to identify, although the main structure remains visible.

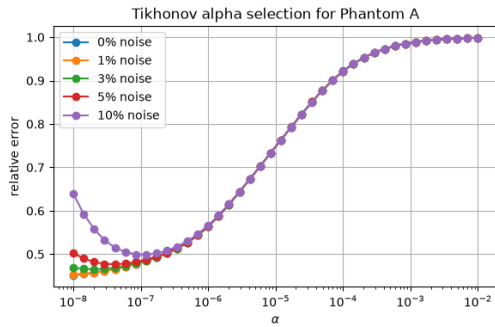


Figure 7: Relative reconstruction error as a function of the Tikhonov regularization parameter α for Phantom A under different noise levels.

Figure 7 shows the relative reconstruction error as a function of the Tikhonov parameter α . The curves reflect the trade-off in $(J^T J + \alpha I)z = J^T \Delta V^\delta$ between noise amplification for small α and regularization bias for large α . The optimal parameter lies consistently between 10^{-8} and 10^{-7} , with the minimum error increasing as the noise level increases. For $\alpha \gtrsim 10^{-6}$, the curves nearly coincide, indicating that the reconstruction is dominated by the smoothness prior rather than the data. Since Phantom A contains a sharp conductivity interface, this behaviour highlights the inherent mismatch between the quadratic regularization model and the true conductivity, implying that a substantial reconstruction bias remains even under optimal parameter selection.

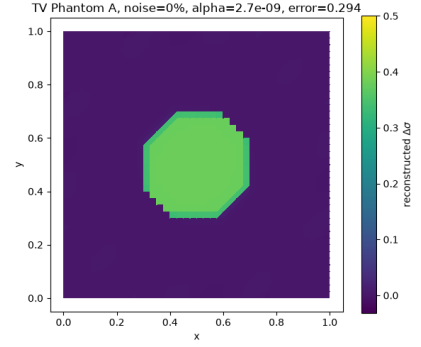


Figure 8: TV reconstruction of Phantom A for noise level 0%.

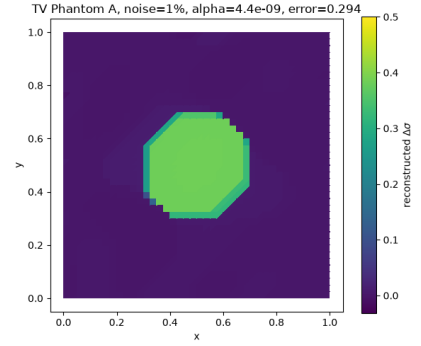


Figure 9: TV reconstruction of Phantom A for noise level 1%.

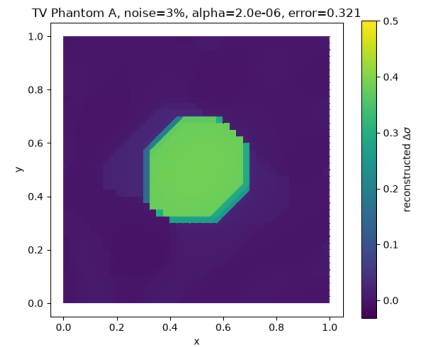


Figure 10: TV reconstruction of Phantom A for noise level 3%.

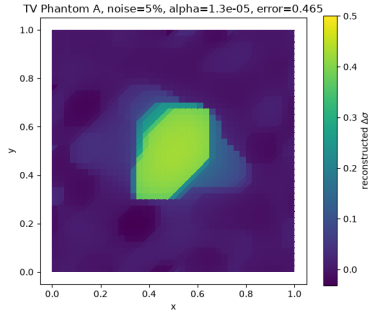


Figure 11: TV reconstruction of Phantom A for noise level 5%.

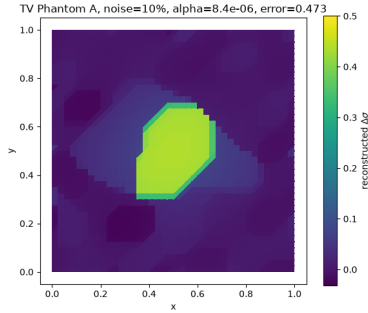


Figure 12: TV reconstruction of Phantom A for noise level 10%.

Figures 8–12 show that TV regularization preserves the conductivity interface and recovers the inclusion with high contrast across all noise levels. This behaviour is consistent with the TV penalty $TV(z)$, which favours piecewise-constant solutions while permitting discontinuities. As the noise level increases, the recovered inclusion becomes increasingly polygonal, indicating that the reconstruction is influenced more strongly by the regularization model and mesh discretization than by the underlying circular geometry.

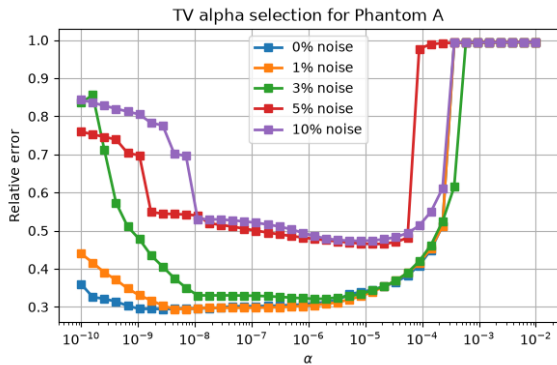


Figure 13: Relative reconstruction error as a function of the TV regularization parameter α for Phantom A under different noise levels.

Figure 13 shows the relative reconstruction error as a function of the TV regularization parameter α for different noise levels. Unlike the Tikhonov results, the error curves exhibit a broad near-optimal region from 10^{-8} to

10^{-5} rather than a sharply defined minimum, indicating that TV regularization is comparatively insensitive to the precise choice of α . This behaviour reflects the ability of the TV penalty to preserve the dominant discontinuity while suppressing oscillatory artefacts over a wide range of parameter values. As the noise level increases, the minimum achievable error increases and the near-optimal region shifts slightly upward, reflecting the loss of recoverable information in the measurements. For sufficiently large α , the error rises rapidly towards unity as the regularization term overwhelms the data fidelity term, forcing excessively simplified piecewise-constant reconstructions. The broad minima and relatively low reconstruction errors observed across several orders of magnitude of α demonstrate the robustness of TV regularization for recovering discontinuous conductivity distributions.

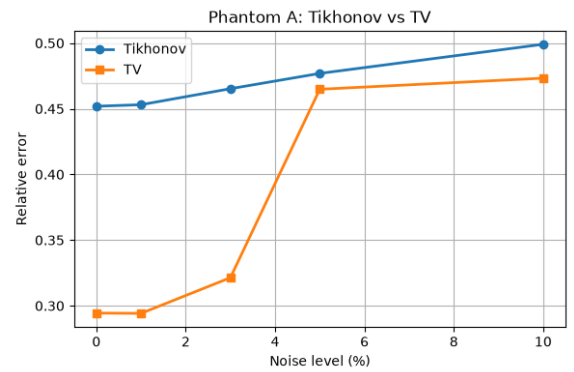


Figure 14: Relative reconstruction error for Tikhonov and TV regularization applied to Phantom A under increasing noise levels.

Figure 14 compares the minimum reconstruction errors obtained by Tikhonov and TV regularization for Phantom A. The results support the hypothesis that TV regularization is more suitable for discontinuous conductivity distributions, yielding lower reconstruction error than Tikhonov across all noise levels considered. Since Phantom A consists of two nearly constant regions separated by a sharp interface, the conductivity is naturally compatible with the TV objective

$$\frac{1}{2} \|Jz - \Delta V^\delta\|_2^2 + \alpha TV(z),$$

which preserves discontinuities while suppressing oscillatory artefacts. In contrast, the Tikhonov solution

$$(J^T J + \alpha I)z = J^T d$$

penalizes large variations and therefore smooths the conductivity jump, introducing a systematic bias even in the absence of measurement noise.

It is noteworthy that both methods exhibit substantial reconstruction error at 0% noise, indicating that modelling error remains significant even for perfect data. Since the reconstruction is based on the linearized model

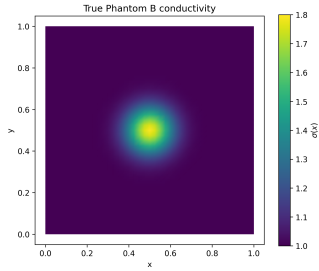
$$F(\sigma_0 + \Delta\sigma) = F(\sigma_0) + J\Delta\sigma + O(\|\Delta\sigma\|^2),$$

together with a finite-dimensional piecewise-constant conductivity representation, the true conductivity cannot be

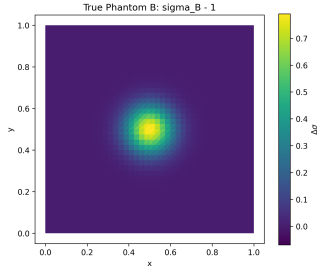
recovered exactly even when $\Delta V^\delta = \Delta V$. The nonzero baseline error therefore reflects the combined effects of linearization, discretization and regularization bias rather than measurement noise alone.

As the noise level increases, the Tikhonov error grows gradually, whereas the TV error remains comparatively low up to approximately 3% noise before increasing sharply between 3% and 5%. This behaviour suggests a transition from a data-dominated regime, in which the conductivity interface is accurately localized by the measurements, to a regularization-dominated regime, where uncertainty in the interface location becomes the dominant source of error. Since the reconstruction error is governed primarily by geometric errors in the recovered boundary rather than smooth amplitude errors, relatively small interface displacements can produce comparatively large increases in the relative error.

4.2 Phantom B



(a) True conductivity σ_B .



(b) Conductivity perturbation $\Delta\sigma_B = \sigma_B - \sigma_0$.

Figure 15: Ground-truth conductivity distribution and conductivity perturbation for Phantom B.

Figure 15 shows the ground-truth conductivity and conductivity perturbation for Phantom B. Since the inverse problem reconstructs $\Delta\sigma_B = \sigma_B - \sigma_0$, the perturbation image in Figure 15b is used as the reference for all subsequent reconstruction comparisons.

We first examine the reconstructions obtained using Tikhonov regularization. Figures 16–20 show the reconstructed conductivity perturbations for Phantom B under increasing noise levels.

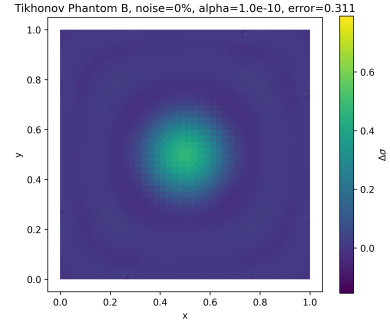


Figure 16: Tikhonov reconstruction of Phantom B for noise level 0%.

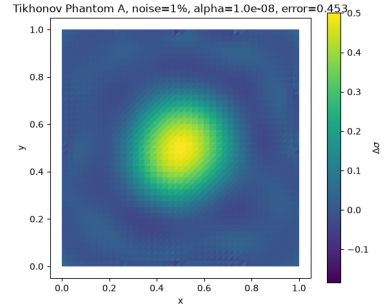


Figure 17: Tikhonov reconstruction of Phantom B for noise level 1%.

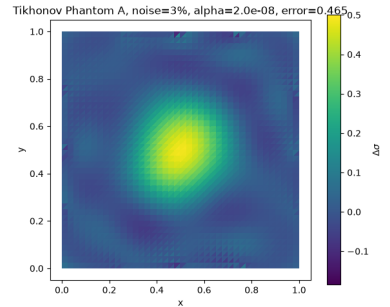


Figure 18: Tikhonov reconstruction of Phantom B for noise level 3%.

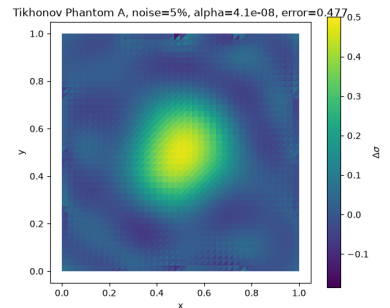


Figure 19: Tikhonov reconstruction of Phantom B for noise level 5%.

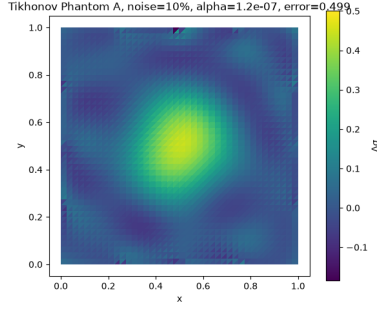


Figure 20: Tikhonov reconstruction of Phantom B for noise level 10%.

Figures 16–20 show that Tikhonov regularization preserves the smooth Gaussian structure of Phantom B across all noise levels. While increasing noise introduces background artefacts and raises the reconstruction error from approximately 0.31 to 0.50, the location and overall shape of the inclusion remain largely unchanged. This is consistent with the smoothness assumptions imposed by the quadratic regularization term.

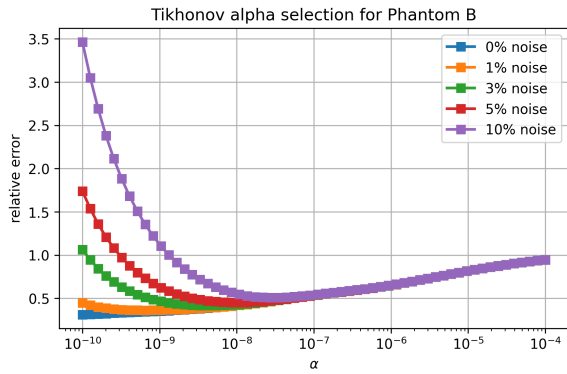


Figure 21: Relative reconstruction error as a function of the Tikhonov regularization parameter α for Phantom B under different noise levels.

Figure 21 exhibits the characteristic U-shaped behaviour of Tikhonov regularization. For small α , the solution of $(J^T J + \alpha I)z = J^T \Delta V^\delta$ is strongly affected by the ill-conditioning of J , leading to noise amplification and large reconstruction errors. Increasing α suppresses these unstable components, while excessively large values introduce increasing regularization bias and oversmooth the solution. The optimal parameter shifts towards larger values as the noise level increases, reflecting the greater regularization required for noisier data. The substantial error reduction achieved through parameter tuning suggests that, for Phantom B, reconstruction accuracy is limited primarily by instability of the inverse problem rather than incompatibility between the regularization model and the underlying conductivity structure.

We next examine the reconstructions obtained using TV regularization. Figures 22–26 show the reconstructed conductivity perturbations for Phantom B under increasing noise levels.

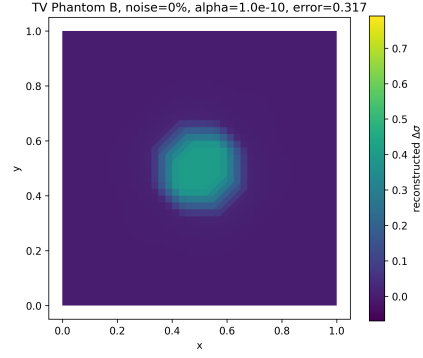


Figure 22: TV reconstruction of Phantom B for noise level 0%.

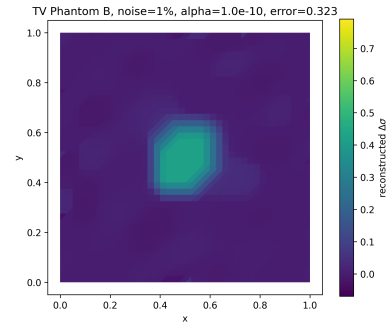


Figure 23: TV reconstruction of Phantom B for noise level 1%.

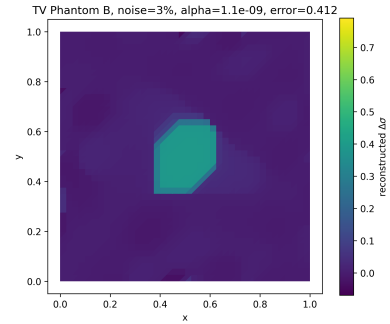


Figure 24: TV reconstruction of Phantom B for noise level 3%.

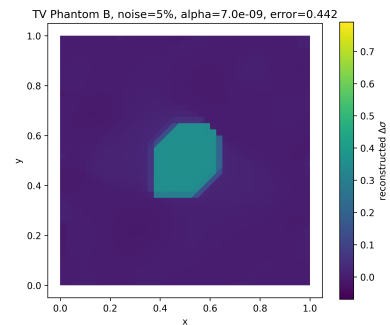


Figure 25: TV reconstruction of Phantom B for noise level 5%.

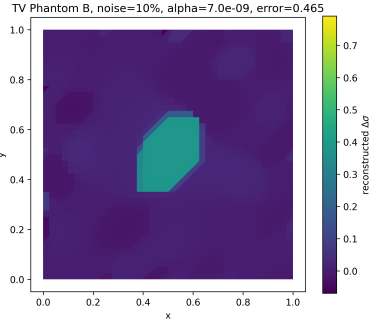


Figure 26: TV reconstruction of Phantom B for noise level 10%.

Figures 22–26 show that TV regularization recovers the location and contrast of the inclusion but not its smooth Gaussian structure. The reconstructions are increasingly characterized by piecewise-constant regions and polygonal boundaries, reflecting the tendency of $TV(z)$ to concentrate variation on a small number of interfaces. Consequently, although TV effectively suppresses oscillatory artefacts, it introduces a structural bias when the true conductivity varies smoothly.

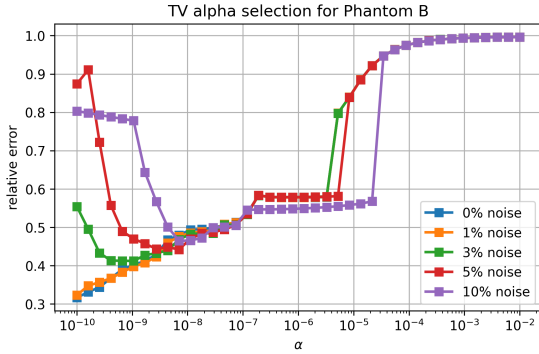


Figure 27: Relative reconstruction error as a function of the TV regularization parameter α for Phantom B under different noise levels.

Figure 27 shows the relative error for TV regularization applied to Phantom B. Unlike the smooth U-shaped curves observed for Tikhonov regularization, the error exhibits distinct regimes reflecting the nonsmooth objective $\frac{1}{2}\|Jz - \Delta V^\delta\|_2^2 + \alpha TV(z)$. For very small α , the reconstruction is under-regularized and highly sensitive to both noise and the ill-conditioning of J . As α increases to approximately 10^{-8} – 10^{-6} , the curves enter a broad plateau and become nearly independent of noise, indicating that noise amplification has been suppressed and the dominant error is instead due to the mismatch between the piecewise-constant TV prior and the smooth Gaussian conductivity. A further increase to $\alpha \approx 10^{-5}$ – 10^{-4} produces a sharp rise in error, corresponding to a qualitative change in the minimizer as the TV penalty overwhelms the data fidelity term and drives the reconstruction towards an overly simplified piecewise-constant structure. These distinct regimes illustrate the fundamentally different parameter dependence of TV and quadratic regularization.

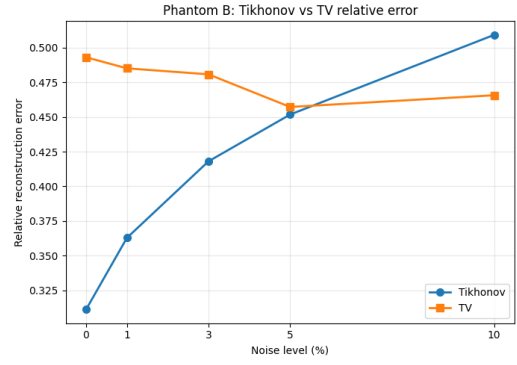


Figure 28: Minimum reconstruction error obtained by Tikhonov and TV regularization for Phantom B when the TV parameter search is restricted to $\alpha \in [10^{-7}, 10^{-4}]$.

Figure 28 compares the minimum reconstruction errors obtained by Tikhonov and TV regularization for Phantom B when the TV parameter search is restricted to $\alpha \in [10^{-7}, 10^{-4}]$, ensuring that the regularization term has a genuine influence on the reconstruction. Tikhonov achieves lower reconstruction errors over low to moderate noise levels (0–5%), while TV becomes marginally more accurate at the highest noise level of 10%. This behaviour is broadly consistent with the original hypothesis. Phantom B consists of a smooth Gaussian conductivity distribution, which is naturally compatible with the quadratic penalty

$$\frac{1}{2}\|Jz - \Delta V^\delta\|_2^2 + \alpha\|z\|_2^2,$$

since Tikhonov regularization promotes smooth solutions while suppressing noise amplification. In contrast, the TV objective

$$\frac{1}{2}\|Jz - \Delta V^\delta\|_2^2 + \alpha TV(z)$$

favours piecewise-constant reconstructions, introducing a structural bias that is incompatible with the smooth Gaussian phantom. The slight advantage of TV at the 10% noise level suggests that, under sufficiently noisy measurements, its stronger noise suppression can outweigh this structural mismatch. Overall, the results indicate that reconstruction quality is governed by both the compatibility between the regularization prior and the underlying conductivity structure and the level of measurement noise. For smooth conductivity distributions, Tikhonov is generally the more appropriate choice, whereas TV is more suitable for recovering discontinuous conductivities, as demonstrated previously for Phantom A.

4.3 Fundamental Modelling Limitations

A notable feature of both Phantom A and Phantom B is the presence of substantial reconstruction error even in the absence of measurement noise. Since the minimum relative errors remain significantly above zero for both regularization methods, the dominant limitation cannot be attributed solely to data contamination. Instead, the results indicate the presence of systematic modelling errors inherent in the linearized EIT framework.

The first source of error arises from the linearization of the

nonlinear forward map. The synthetic measurements are generated from

$$\Delta V = F(\sigma) - F(\sigma_0),$$

whereas the reconstruction is based on the first-order approximation

$$F(\sigma_0 + \Delta\sigma) = F(\sigma_0) + J\Delta\sigma + \mathcal{O}(\|\Delta\sigma\|^2).$$

Even for noise-free data, the neglected higher-order term generally remains nonzero, so the exact conductivity perturbation does not satisfy the linearized model exactly. Consequently, part of the reconstruction error is already present before regularization is applied.

A second source of error is discretization. The conductivity perturbation is represented on the reconstruction mesh by

$$\Delta\sigma|_{q_n} = z_n,$$

so the inverse problem is solved in a finite-dimensional space of piecewise-constant functions. Smooth conductivity distributions, such as Phantom B, cannot be represented exactly within this space, introducing an additional approximation error even if the linearized model were exact. Finally, regularization introduces an unavoidable bias. For Tikhonov regularization, the reconstruction satisfies

$$(J^T J + \alpha I)z = J^T \Delta V^\delta,$$

while TV regularization solves

$$\min_z \left\{ \frac{1}{2} \|Jz - \Delta V^\delta\|_2^2 + \alpha \text{TV}(z) \right\}.$$

In both cases, any $\alpha > 0$ shifts the solution away from the unregularized inverse problem towards functions preferred by the chosen prior. This bias is beneficial for stability but necessarily introduces reconstruction error.

Taken together, the numerical results suggest that the total reconstruction error can be viewed qualitatively as the combined effect of linearization error, discretization error, regularization bias, and measurement noise. The persistence of substantial error at 0% noise indicates that, for the conductivity contrasts considered here, modelling error is at least as significant as measurement noise. This highlights a fundamental limitation of linearized EIT: reconstruction accuracy is constrained not only by data quality, but also by the validity of the mathematical model used to represent the inverse problem.

5 CONCLUSION

The numerical experiments demonstrate that the effectiveness of a regularization method in EIT depends primarily on the compatibility between its implicit prior and the underlying conductivity structure. TV regularization was more effective for the discontinuous Phantom A, while Tikhonov regularization was generally more suitable for the smooth Phantom B. The results further show that reconstruction error is not determined solely by measurement noise; even in the absence of noise, significant errors remain due to linearization, discretization, and regularization bias. These findings highlight both the importance of prior selection in EIT reconstruction and the intrinsic limitations of the linearized inverse problem.

REFERENCES

- [1] A.-P. Calderón, “On an Inverse Boundary Value Problem,” in *Seminar on Numerical Analysis and its Applications to Continuum Physics*, Available at: <https://sites.math.washington.edu/~reu/papers/reference/calderon.pdf>
- [2] A. N. Tikhonov and V. Y. Arsenin, *Solutions of Ill-Posed Problems*, Winston & Sons, Washington, DC, 1977.
Available at: <https://www.scirp.org/reference/referencespapers?referenceid=3161072>
- [3] L. I. Rudin, S. Osher, and E. Fatemi, “Non-linear Total Variation Based Noise Removal Algorithms,” *Physica D: Nonlinear Phenomena*, Available at: <https://www.sciencedirect.com/science/article/pii/016727899290242F>
- [4] A. Borsic, B. M. Graham, A. Adler, and W. R. B. Lionheart, “Total Variation Regularization in Electrical Impedance Tomography,” *IEEE Transactions on Medical Imaging*, Available at: <https://www.sce.carleton.ca/faculty/adler/publications/2007/borsic-2007-TV-EIT.pdf>
- [5] Y. Fan and L. Ying, “Solving Electrical Impedance Tomography with Deep Learning,” *Journal of Computational Physics*, vol. 404, 109119, 2020.
Available at: <https://doi.org/10.1016/j.jcp.2019.109119>
- [6] B. Gong, B. Schullcke, S. Krueger-Ziolek, and K. Moeller, “EIT Reconstruction Using Higher Order TV Regularization,” *IFAC PapersOnLine*, Available at: <https://doi.org/10.1016/j.ifacol.2017.08.1646>
- [7] D. S. Holder (ed.), *Electrical Impedance Tomography: Methods, History and Applications*, Institute of Physics Publishing, Bristol, 2005.
- [8] A. Adler and W. R. B. Lionheart, “Uses and abuses of EIDORS: An extensible software base for EIT,” *Physiological Measurement*, 2006.
Available at: <https://doi.org/10.1088/0967-3334/27/5/S03>
- [9] Y. Saad, *Iterative Methods for Sparse Linear Systems*, 2nd ed., <https://www.stat.uchicago.edu/~lekheng/courses/324/saad.pdf>
- [10] H. W. Engl, M. Hanke, and A. Neubauer, *Regularization of Inverse Problems*,
Available at: <https://link.springer.com/book/9780792341574>